AUTOSAR High-Level Design Specification

1. Document Metadata & System Overview

AUTOSAR HIGH-LEVEL DESIGN (HLD) SPECIFICATION
DOCUMENT CLASSIFICATION: SYNTHETIC DEMONSTRATION DATA FOR TECHNICAL EVALUATION ONLY.
NOTICE: This document is an illustrative, synthetic engineering benchmark designed for technical evaluation of automated analysis systems. It is NOT an OEM-approved production architecture.

1.1 System Architecture Scope
The Powertrain and Chassis Domain Controller (PCDC) coordinates electric traction, friction braking, energy recuperation, and high-voltage power distribution for a dual-motor electric vehicle. The architecture conforms to the AUTOSAR Classic Platform Release 4.4.0 standards.

1.2 System Level Goals
- Coordinate brake blending between friction braking and electric regenerative braking.
- Maintain vehicle lateral and longitudinal stability under emergency deceleration.
- Manage high-voltage battery state of charge (SoC) and cell temperature derating.
- Comply with ISO 26262 ASIL-D functional safety requirements across braking and torque intervention loops.

Software Architecture Overview

2. AUTOSAR Software Layering & Virtual Functional Bus

2.1 Virtual Functional Bus (VFB) Representation
The PCDC application layer resides above the AUTOSAR Run-Time Environment (RTE). All inter-component interactions are mediated via well-defined Port Interfaces according to AUTOSAR Virtual Functional Bus specifications.

2.2 Software Component (SWC) Inventory
The system architecture defines five primary application and service software components:
1. VehicleSpeedSWC: Sensor abstraction and vehicle velocity calculation component.
2. BrakeControlSWC: Core anti-lock braking (ABS), electronic stability program (ESP), and brake arbitration component.
3. MotorControlSWC: Electric traction inverter torque command generator and regenerative power manager.
4. BatteryManagementSWC: High-voltage battery pack state estimation, thermal protection, and current derating monitor.
5. DiagnosticManagerSWC: On-board diagnostics (OBD-II / UDS ISO 14229) service component, fault code storage, and fail-safe coordinator.

Software Component: VehicleSpeedSWC

3. Software Component Specification: VehicleSpeedSWC

3.1 Component Identification
- Component Name: VehicleSpeedSWC
- Category: Sensor Actuator Software Component (SensorSWC)
- Safety Integrity Level: ASIL-B
- Execution Periodicity: 10ms cyclic runnable (Runnable_CalculateSpeed)

3.2 Functional Description & Responsibilities
VehicleSpeedSWC interfaces with four wheel speed pulse counters via Complex Device Drivers (CDD). It is responsible for:
- Reading raw wheel rotation pulse rates from front-left, front-right, rear-left, and rear-right sensors.
- Filtering high-frequency wheel slip noise using a recursive Kalman filter.
- Calculating reference vehicle linear speed in kilometers per hour (kph).
- Detecting wheel speed sensor plausibility faults and reporting diagnostic events.

3.3 Associated Interfaces
- Provided Interface: If_VehicleSpeed (Sender-Receiver)
- Required Interface: If_WheelPulseCDD (Client-Server)
- Required Interface: If_DiagnosticService (Client-Server)

Software Component: BrakeControlSWC

4. Software Component Specification: BrakeControlSWC

4.1 Component Identification
- Component Name: BrakeControlSWC
- Category: Application Software Component (ApplicationSWC)
- Safety Integrity Level: ASIL-D
- Execution Periodicity: 10ms cyclic runnable (Runnable_BrakeArbitration)

4.2 Functional Description & Responsibilities
BrakeControlSWC is the primary deceleration authority in the chassis domain. Its core duties include:
- Acquiring driver brake pedal travel percentage and hydraulic line pressure.
- Computing total target vehicle braking deceleration torque.
- Determining brake blending ratio: allocating maximum feasible regenerative braking to MotorControlSWC while applying remainder via electro-hydraulic friction brakes.
- Triggering Anti-Lock Braking (ABS) modulation when individual wheel slip exceeds 15 percent.

4.3 Associated Interfaces & Dependencies
- BrakeControlSWC depends on VehicleSpeedSWC for vehicle velocity and individual wheel speeds.
- Provided Interface: If_BrakeCommand (Sender-Receiver)
- Provided Interface: If_BrakeTorqueCommand (Sender-Receiver)
- Required Interface: If_VehicleSpeed (Sender-Receiver)
- Required Interface: If_DiagnosticService (Client-Server)

Software Component: MotorControlSWC

5. Software Component Specification: MotorControlSWC

5.1 Component Identification
- Component Name: MotorControlSWC
- Category: Application Software Component (ApplicationSWC)
- Safety Integrity Level: ASIL-D
- Execution Periodicity: 5ms cyclic runnable (Runnable_MotorTorqueControl)

5.2 Functional Description & Responsibilities
MotorControlSWC translates acceleration pedal inputs and brake blending requests into front and rear inverter torque commands.
- Coordinates electric motor traction torque delivery up to 450 Nm peak torque.
- Executes regenerative braking deceleration torque requests requested by BrakeControlSWC.
- Limits instantaneous torque output based on maximum allowable discharge and charge current limits from BatteryManagementSWC.
- MotorControlSWC depends on BatteryManagementSWC for high-voltage battery state of charge and thermal power limits.

5.3 Associated Interfaces
- Provided Interface: If_MotorTorque (Sender-Receiver)
- Required Interface: If_BrakeTorqueCommand (Sender-Receiver)
- Required Interface: If_BatteryState (Sender-Receiver)
- Required Interface: If_PowerDerateRequest (Client-Server)

Software Component: BatteryManagementSWC

6. Software Component Specification: BatteryManagementSWC

6.1 Component Identification
- Component Name: BatteryManagementSWC
- Category: Application Software Component (ApplicationSWC)
- Safety Integrity Level: ASIL-C
- Execution Periodicity: 50ms cyclic runnable (Runnable_BatteryMonitor)

6.2 Functional Description & Responsibilities
BatteryManagementSWC oversees the 800V lithium-ion traction battery pack:
- Computes state of charge (SoC), state of health (SoH), and internal cell resistance.
- Monitors cell voltages (minimum 3.0V, maximum 4.2V) and multi-point pack temperature probes.
- Issues power derating warnings to MotorControlSWC when pack temperature exceeds 55 degrees Celsius.
- Controls main high-voltage pyrotechnic and electromechanical contactor isolation in emergency cutoff scenarios.

6.3 Associated Interfaces
- Provided Interface: If_BatteryState (Sender-Receiver)
- Provided Interface: If_PowerDerateRequest (Client-Server)
- Required Interface: If_DiagnosticService (Client-Server)

Software Component: DiagnosticManagerSWC

7. Software Component Specification: DiagnosticManagerSWC

7.1 Component Identification
- Component Name: DiagnosticManagerSWC
- Category: Service Component / Complex Driver (ServiceSWC)
- Safety Integrity Level: ASIL-B
- Execution Periodicity: 100ms background runnable (Runnable_DiagnosticService)

7.2 Functional Description & Responsibilities
DiagnosticManagerSWC provides centralized diagnostic event logging, fault debouncing, and UDS ISO 14229 compliance:
- Collects diagnostic trouble codes (DTCs) from BrakeControlSWC, MotorControlSWC, and BatteryManagementSWC.
- Implements Client-Server interface If_DiagnosticService with operations SetDTCStatus(), GetFaultReport(), and ClearDTC().
- Freezes environmental freeze-frame telemetry upon detection of confirmed safety faults.
- Manages vehicle limp-home and fail-safe transitions when critical sensor loss occurs.

7.3 Associated Interfaces
- Provided Interface: If_DiagnosticService (Client-Server)
- Required Interface: If_VehicleSpeed (Sender-Receiver)

Port Interfaces Specification

8. Detailed Port Interfaces & Communication Matrix

8.1 Sender-Receiver Port Interfaces
The following Sender-Receiver interfaces provide asynchronous data communication across the AUTOSAR RTE:

Interface: If_VehicleSpeed
- Interface Type: Sender-Receiver
- Direction: Provided by VehicleSpeedSWC, Required by BrakeControlSWC and DiagnosticManagerSWC
- Data Elements:
* Sig_VehicleSpeed_kph (float32, 0.0 to 300.0 kph)
* Sig_WheelSpeed_FL_rpm (uint16, 0 to 4000 rpm)
* Sig_WheelSpeed_FR_rpm (uint16, 0 to 4000 rpm)

Interface: If_BrakeTorqueCommand
- Interface Type: Sender-Receiver
- Direction: Provided by BrakeControlSWC, Required by MotorControlSWC
- Data Elements:
* Sig_TargetBrakeTorque_Nm (float32, -2000.0 to 0.0 Nm)
* Sig_BrakePedalPosition_pct (uint8, 0 to 100 %)

Interface: If_MotorTorque
- Interface Type: Sender-Receiver
- Direction: Provided by MotorControlSWC, Required by BrakeControlSWC
- Data Elements:
* Sig_ActualMotorTorque_Nm (float32, -2000.0 to +2000.0 Nm)
* Sig_MotorTemp_degC (int16, -40 to +180 degC)

Interface: If_BatteryState
- Interface Type: Sender-Receiver
- Direction: Provided by BatteryManagementSWC, Required by MotorControlSWC
- Data Elements:
* Sig_BatterySoC_pct (uint8, 0 to 100 %)
* Sig_BatteryCurrent_A (float32, -600.0 to +600.0 A)
* Sig_BatteryVoltage_V (float32, 500.0 to 900.0 V)

Client-Server Interfaces Specification

9. Client-Server Operations & Service Interfaces

9.1 Client-Server Interfaces
Synchronous and client-server request-response operations are modeled via dedicated Client-Server interfaces:

Interface: If_DiagnosticService
- Interface Type: Client-Server
- Direction: Provided by DiagnosticManagerSWC, Required by BrakeControlSWC, MotorControlSWC, and BatteryManagementSWC
- Operations:
* SetDTCStatus(in uint32 DTCNumber, in uint8 FaultStatus) -> uint8 ReturnCode
* GetFaultReport(in uint32 DTCNumber, out FaultRecordStructure FaultRecord) -> uint8 ReturnCode
* ClearDTC(in uint32 DTCGroup) -> uint8 ReturnCode

Interface: If_PowerDerateRequest
- Interface Type: Client-Server
- Direction: Provided by BatteryManagementSWC, Required by MotorControlSWC
- Operations:
* RequestTorqueLimit(in float32 MaxAllowableTorque_Nm, in uint8 ReasonCode) -> boolean Acknowledged
* GetCurrentDeratingStatus(out uint8 DerateState) -> void

Signals and Data Elements Catalog

10. Signals and Data Dictionary

10.1 Global Signal Data Dictionary
The table below documents all active signals exchanged across the RTE:

Signal Name	Sender SWC	Receiver SWC	Data Type	Units	Range
Sig_VehicleSpeed_kph	VehicleSpeedSWC	BrakeControlSWC	float32	kph	0.0 - 300.0
Sig_WheelSpeed_FL_rpm	VehicleSpeedSWC	BrakeControlSWC	uint16	rpm	0 - 4000
Sig_WheelSpeed_FR_rpm	VehicleSpeedSWC	BrakeControlSWC	uint16	rpm	0 - 4000
Sig_TargetBrakeTorque_Nm	BrakeControlSWC	MotorControlSWC	float32	Nm	-2000.0 - 0.0
Sig_BrakePedalPosition_pct	BrakeControlSWC	MotorControlSWC	uint8	%	0 - 100
Sig_ActualMotorTorque_Nm	MotorControlSWC	BrakeControlSWC	float32	Nm	-2000.0 - 2000.0
Sig_MotorTemp_degC	MotorControlSWC	DiagnosticManagerSWC	int16	degC	-40 - 180
Sig_BatterySoC_pct	BatteryManagementSWC	MotorControlSWC	uint8	%	0 - 100
Sig_BatteryCurrent_A	BatteryManagementSWC	MotorControlSWC	float32	A	-600.0 - 600.0
Sig_BatteryVoltage_V	BatteryManagementSWC	MotorControlSWC	float32	V	500.0 - 900.0

Architectural Dependencies & Functional Flows

11. Component Dependencies and Run-Time Interactions

11.1 Component Architectural Dependencies
The verified dependencies across software components are summarized below:
- BrakeControlSWC depends on VehicleSpeedSWC for accurate ground speed reference and anti-lock slip control.
- MotorControlSWC depends on BrakeControlSWC for regenerative braking torque setpoint coordination.
- MotorControlSWC depends on BatteryManagementSWC for maximum battery charge current and thermal derating limits.
- BrakeControlSWC calls DiagnosticManagerSWC to report ABS solenoid faults and pressure plausibility errors.
- BatteryManagementSWC calls DiagnosticManagerSWC to register cell over-voltage, thermal runaway, and insulation breakdown events.
- MotorControlSWC calls DiagnosticManagerSWC to report inverter over-temperature and phase short-circuit events.

11.2 Regenerative Braking Coordination Functional Flow
1. Driver depresses brake pedal. BrakeControlSWC samples pedal stroke via 10ms runnable.
2. BrakeControlSWC queries VehicleSpeedSWC to check if vehicle speed exceeds 5 kph.
3. BrakeControlSWC queries MotorControlSWC for available regen torque capacity.
4. If BatteryManagementSWC allows charging (SoC < 98% and pack temp < 50C), BrakeControlSWC commands negative regenerative torque to MotorControlSWC via If_BrakeTorqueCommand.
5. Remaining deceleration demand is fulfilled by the friction braking calipers.

System Integration & Safety Requirements

12. Integration, RTE Mapping, and ISO 26262 Requirements

12.1 RTE Task Scheduling & Task Allocation
- 5ms Periodic Task (OS_Task_5ms, Priority 30): Hosts MotorControlSWC Runnable_MotorTorqueControl.
- 10ms Periodic Task (OS_Task_10ms, Priority 25): Hosts VehicleSpeedSWC Runnable_CalculateSpeed and BrakeControlSWC Runnable_BrakeArbitration.
- 50ms Periodic Task (OS_Task_50ms, Priority 15): Hosts BatteryManagementSWC Runnable_BatteryMonitor.
- 100ms Periodic Task (OS_Task_100ms, Priority 10): Hosts DiagnosticManagerSWC Runnable_DiagnosticService.

12.2 Functional Safety & ISO 26262 Compliance
- BrakeControlSWC and MotorControlSWC are classified as ASIL-D safety items. Freedom from interference (FFI) is enforced via hardware Memory Protection Unit (MPU) partitioning.
- BatteryManagementSWC is classified as ASIL-C.
- VehicleSpeedSWC and DiagnosticManagerSWC are classified as ASIL-B.
- Watchdog Manager (WdgM) performs alive supervision on 5ms and 10ms tasks with a maximum fault reaction tolerance time (FRTT) of 20 milliseconds.